

Acne

Dr eman gomaa

Consultant dermatologist

-
- acne vulgaris is a multifactorial disorder of the pilosebaceous unit. The clinical picture can vary significantly, from mild comedonal acne to fulminant systemic disease.



Fig. 36.6 Moderate to severe acne vulgaris. Multiple coalescing papules, pustules, and small nodules are present on the cheek. Courtesy, Kalman Watsky, MD.

Aetiopathogenesis

4 stages

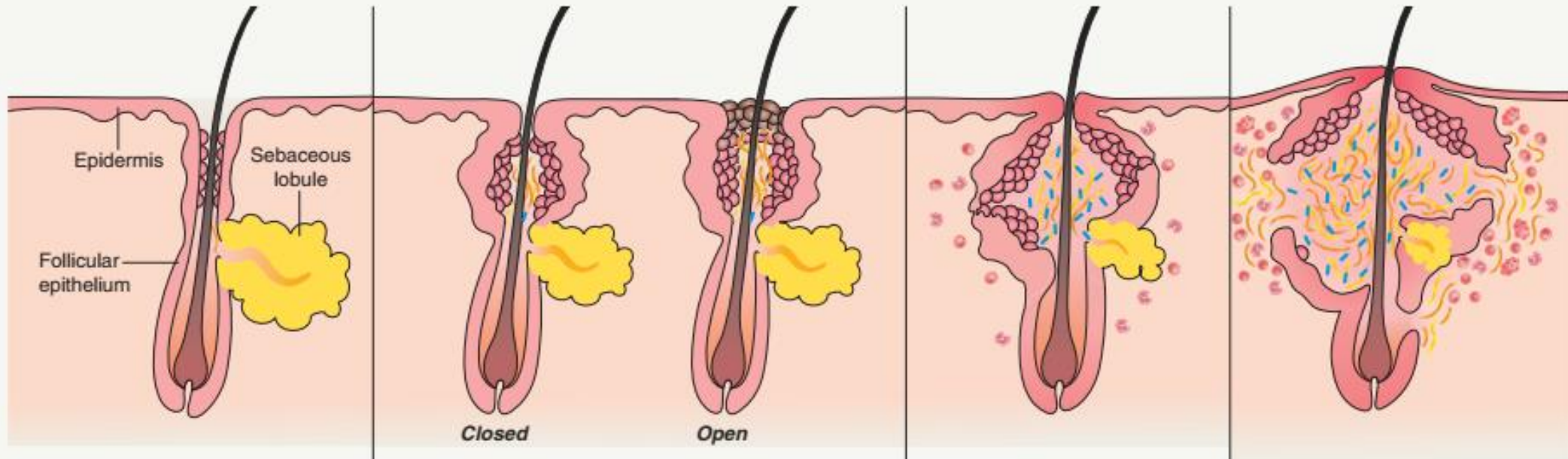
Increase sebum production

Ductal hypercornification

Proliferation of *Propionibacterium acne*

Inflammation

PATHOGENESIS OF ACNE



Early comedo

- Hyperkeratosis and $\uparrow$ comeocyte cohesiveness in the upper sebaceous follicle, which lead to microcomedo formation
- Androgen stimulation of sebum production

Later comedo

- Accumulation of shed keratin and sebum
- Formation of whorled lamellar concretions
- Comedo may be *closed* (no obvious follicular opening) or *open* (dilated follicular opening; keratin plug darkens due to oxidized lipids & melanin)

Inflammatory papule/pustule

- *Propionibacterium acnes* proliferation, which upregulates innate immune responses (e.g. via TLRs)
- Mild inflammation (primarily neutrophils), which increases upon rupture of the comedo wall
- Sebaceous lobule regression

Nodule/cyst

- Marked inflammation (primarily T cells)
- May lead to scarring

aetiopathogenesis

Sebum production	Ductal hypercornification	Proleferation of Propionibacterium acne	Inflammation
Level of sebum correlate with severity • Androgenic stimulation at time of puberty • Due to end organ hyperresponce of gland to normal level of androgen • No increase androgen level in most pts	• Due to excessive accumulation of ductal corneocyte • Due to increase formation or inadequet desquamation • Cause of ductal hypercornifecation 1. Andogen stimulation 2. Sebum irritation	• Due to anaerobic environment caused by ductal hypercornification • P acne →increase lipase→cleave lipid into proinflam- FFA →rupture of bilosebaceous unit • Bind, activate of TLR 2 →(early specific)	• Due to libration of lip enzyme from p.acne →rupture of follicle →inflammation • 2 stages 1. Early specific 2. Late non specific - Ruptur follicle →inflammation

TLR

1] P. acne activate → TLR2 on perifollicular inflammatory cells → increased pro-inflammatory mediator IL-1B-IL-8-TNF alpha → recruitment neutrophils , release lysosome & disruption of follicular epithelium → initiating steps in comedogenesis

N.B:- TLR present on basal-infundibular KCs of sebaceous gland

2] B- defensins:-

pro-inflammatory cytokines → IL-1B & TNF alpha

Bacterial lipopolysaccharide & up regular B-defensins.



Fig. 36.3 Comedonal acne vulgaris. On the cheek (**A**) and forehead (**B**), there are open and closed comedones as well as postinflammatory hyperpigmentation (**A**) and inflammatory papules (**B**). *B, Courtesy, Kalman Watsky, MD.*



Fig. 36.4 Moderate acne vulgaris. Scattered papulopustules and



Fig. 36.5 Moderate to severe acne vulgaris. Inflammatory papules and pustules as well as both open and closed comedones are evident on the cheek, forehead and chin. Note the nodular lesion on the temple and open comedones in the concha of the ear. Scarring is present in the preauricular area.



Fig. 36.7 Severe nodular acne. This form is best treated with low doses of isotretinoin initially, to avoid precipitating a flare.

Other factor in acne vulgaris

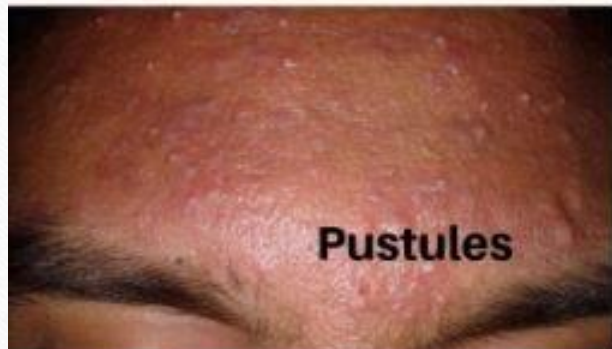
- Genetic:- +ve family history
- Diet:- high glycemic diet & milk → androgen & IGF 1
- Premenstrual factor
- UVR:- improve acne or increase carcinogenicity
- Stress
- Occupational :- hydration → acne / steam & oil
- Sweating → 15% → ductal hydration

Acne C/pic

- Site → face, chest, shoulder & upper back
- Lesion: polymorphic
- Types of comedons:
 - *Open: black
 - *Closed:white *Intermediate *Sandpaper:- multiple variable small → forehead
 - *Macrocomedon *Submarine comedon → large, deep & >0.5 cm
 - *2ry comedon → chloracne , pomad acne & steroid acne
- Progression:- papule, pustule, nodule cyst → erythema, PIH & Scar



eFig. 36.2 Open comedones in a patient with scarring cystic acne.



Variants of acne

Post adolescent acne	Neonatal acne Neonatal cephalic bustulosis	Infantile acne	Acne conglobata	Acne fulminant
Tender deep seated nodules at lower face → jaw line & chin Premenstrual flare	20% at & after birth Male Regress spontaneously Pathogenesis:- Malassezia & androgen from mother Ttt → antifungal	3rd & 6th month May persist till 5 ys Male Adrenal focus secretes androgen & hormonal imbalance	Young male Trunk, face & limb Tender nodule → large cyst → draining sinus [double comedon] Late in life Follicular occlusion triad & tetrad	Like acne conglobate + hemorrhagic nodules → ragged ulcer → gelatinous necrotic depris + systemic symptoms → fever, malaise, leukocytosis, polyarthralgia, EN, HSM Ttt:- steroid + isotretinoin + bed rest & hospitalization



Fig. 36.12 Infantile acne. Presentations can range from numerous open comedones (A) to primarily papulopustules (B). A, Courtesy, Julie V Schaffer, MD; B,

Gram –ve folliculitis	Solid facial edema [Morbihan]	Acne excoriee	Acne mechanica	Endocrinological
Due to prolonged ttt of acne with AB Goove of butules around ant. Narse Ttt:- gram –ve AB + isotretenoin	Disfiguring complication → soft tissue swelling → woody non scaling induration + erythema Ttt:- iso + ketotifen + prednisone	Picking & squeezing of skin Girl 	At site of physical trauma	Hyperandrogenism increase FSH, LH & DHEAS
				

Drug induced acne	Chemical acne	Vasculitic pyoderma gangrenosum acne	
Steroid, androgen & phenytoin Sudden monomorphic papule & pustule	Cosmetic & pomade acne Occupational Chloroacne Detergent Coal tar	Immune reaction to P acne Unresponsive to iso Ttt- steroid / azathioprine	



Fig. 36.14 Drug-induced acne due to isoniazid. Courtesy, Kalma Watsky, MD.

Abert syndrome	Papa syndrome	Sapho syndrome	Hair AN syndrome
Mutation in FGFR2 Acneform papules Seborrhea Nail dystrophy Cutaneous & ocular hypopigment	pyogenic arthritis pyoderma gangreosum Acne	Synovitis Acne Pp Pustulosis Hyperostosis osteitis	Hyperandrogenism Insulin resistance Acanthosis nigricans



DD

DIFFERENTIAL DIAGNOSIS OF ACNE

Acne vulgaris – comedonal

Closed

- Milia
- Osteoma cutis
- Sebaceous hyperplasia
- Pseudoacne of the nasal crease (see Fig. 36.16A)
- Syringomas
- Trichoepitheliomas*
- Trichodiscomas, fibrofolliculomas
- Eruptive vellus hair cysts†, steatocystoma multiplex†
- Colloid milia
- Acne exacerbated by systemic corticosteroids‡ or anabolic steroids
- Contact acne (occupational, pomade, cosmetica, mechanica; chloracne)
- Follicular mucinosis (see Fig. 36.16B)
- Follicular mycosis fungoides (see Fig. 36.16C)

Open

- Contact acne (see Closed)
- Acne exacerbated by systemic corticosteroids‡ or anabolic steroids
- Trichostasis spinulosa (see inset)
- Favre–Racouchot syndrome
- Pseudoacne of the nasal crease (see Fig. 36.16A)
- Nevus comedonicus
- Basaloid follicular hamartoma syndrome
- Familial dyskeratotic comedones
- Dowling–Degos disease
- Radiation-induced comedones
- Molluscum-induced comedones
- Childhood flexural comedones (often a single double-orifice lesion)
- Dilated pore of Winer (single lesion)
- Trichofolliculoma (usually a single lesion)
- Follicular spines – in settings such as trichodysplasia spinulosa§, type VI pityriasis rubra pilaris§, multiple myeloma, demodicosis§, follicular mucinosis, and drug-induced (e.g. BRAF inhibitors, cyclosporine§, acitretin)



Acne vulgaris – inflammatory

- Rosacea
- Perioral/periorificial dermatitis
- Demodicosis (adults > children§)
- Folliculitis – culture-negative (normal flora), staphylococcal, Gram-negative, eosinophilic§, *Pityrosporum*
- Acne/acneiform eruptions due to topical or systemic corticosteroids‡, anabolic steroids, or other medications (e.g. lithium, EGFR inhibitors; see Table 36.1)
- Pseudofolliculitis barbae, acne keloidalis nuchae
- Pseudoacne of the nasal crease
- Furuncle/carbuncle
- Idiopathic facial aseptic granuloma (in children)
- Neutrophilic dermatoses and neutrophilic eccrine hidradenitis
- Keratosis pilaris
- Trichodysplasia spinulosa (viral-associated trichodysplasia)§
- Lupus miliaris disseminatus faciei
- Psychogenic (neurotic) excoriations, factitial lesions
- Follicular mucinosis (see Fig. 36.16B)
- Follicular mycosis fungoides (see Fig. 36.16C)
- Tinea faciei
- Molluscum contagiosum (especially inflamed lesions)
- Angiofibromas
- Dowling–Degos disease (especially the Haber syndrome variant)
- Cutaneous Rosai–Dorfman disease
- Granulomatosis with polyangiitis (Wegener granulomatosis)
- Syndromes associated with acne, eg. PAPA, (PA)PASH, SAPHO, Apert syndrome

Neonatal acne (neonatal cephalic pustulosis)

- Sebaceous hyperplasia
- Milia
- Miliaria rubra (especially pustular variant)
- Candidal infection
- Papulopustular eruption of hyper-IgE syndrome
- Vesiculopustular eruption of transient myeloproliferative disorder

Solid facial edema secondary to acne vulgaris

- Solid facial edema secondary to rosacea
- Melkersson–Rosenthal syndrome, sarcoidosis (e.g. Heerfordt syndrome), lepromatous leprosy
- Lymphoma (B- or T-cell), leukemia cutis, angiosarcoma
- Scleromyxedema, myxedema, self-healing (juvenile) cutaneous mucinosis
- Autoimmune connective tissue disease (e.g. dermatomyositis, lupus erythematosus, Still disease)
- Angioedema (lasts <24–72 hours), C1 esterase inhibitor deficiency (lasts 2–3 days), blepharochalasis/Ascher syndrome
- Sweet syndrome or cellulitis – more abrupt onset
- Superior vena caval obstruction
- Lymphatic malformation
- Pachydermoperiostosis and other genetic disorders with coarse facies
- Other – trichinosis, primary systemic amyloidosis

HISTORY AND PHYSICAL EXAMINATION OF THE ACNE PATIENT

History	Physical examination
Sex	Skin type (e.g. oily versus dry)
Age	Skin color/phototype
Motivation	Distribution of acne
Lifestyle/hobbies	• Face (e.g. "T-zone", cheeks, jawline)
Occupation	• Neck, chest, back, upper arms
Current and previous treatments	Overall degree of involvement (mild, moderate or severe)
Use of cosmetics, sunscreens, cleansers, moisturizers	Lesion morphology
Menstrual history	• Comedones
Medications	• Papules, pustules
• Corticosteroids	• Nodules, cysts
• Oral contraceptives	• Sinus tracts
• Anabolic steroids	Scarring
• Other (e.g. lithium, EGFR inhibitors; see Table 36.1)	• Pitted
Concomitant illnesses	• Hypertrophic
Family history (including severity of acne, polycystic ovary syndrome, inflammatory disease)	• Atrophic
	Postinflammatory pigmentary changes

TREATMENT OF ACNE VULGARIS

	Mild		Moderate		Severe
	Comedonal	Papular/pustular	Papular/pustular	Nodular	Conglobata/fulminans
First line	Topical retinoid	BPO ± topical antibiotic Topical retinoid + topical antimicrobial(s)*	Oral antibiotic† + topical retinoid ± BPO Topical retinoid + BPO ± topical antibiotic	Oral antibiotic† + topical retinoid ± BPO	Oral isotretinoin (may require concurrent use of corticosteroid, esp. in acne fulminans)
Second line	Alternative topical retinoid Azelaic acid Salicylic acid	Alternative topical retinoid and/or topical antimicrobial Azelaic acid Salicylic acid Topical dapsone	Alternative oral antibiotic‡ + alt. topical retinoid ± BPO/azelaic acid	Oral isotretinoin Alternative oral antibiotic‡ + alt. topical retinoid ± BPO/azelaic acid	Oral antibiotic (± high dose) + topical retinoid + BPO Oral dapsone
Options for female patients			Oral contraceptive/antiandrogen	Oral contraceptive/antiandrogen	Oral contraceptive/antiandrogen
Procedural options	Comedo extraction		Comedo extraction	Comedo extraction Intralesional corticosteroid	Intralesional corticosteroid
Refractory to treatment		Exclude Gram-negative folliculitis	Exclude Gram-negative folliculitis		
			Female patient: exclude adrenal or ovarian dysfunction Exclude use of anabolic steroid or other acne-exacerbating medications		
Maintenance			Topical retinoid ± BPO	Topical retinoid ± BPO	Topical retinoid ± BPO

Treatment of acne

topical

systemic

Topical ttt

Topical retinoids

1] tretinoin 1st generation

0.025 – 0.05 %

Mechanism :-

decrease cohescence of
abnormal KCs

Prevent new lesion

Increase penetration of other
therapies

2] adapalin (differin gel) less
irritant more effective

3] tazarotene

Azelaic acid

[skenorine]

Drived from

pityrosporum oval

1] normalize
keratization

2] antimicrobial

3] decrease
tyrosinase enzyme
→ bleaching

Benzyl peroxide
Bacteriostatic
Comedolytic
Anti-inflammatory

Topical AB
Clindamycine
Dalacin C
Erythromycin
acnemycin

Steroid
Topical dapson
sulphur

Systemic anti-acne

Systemic antibiotic

- 1) Tetracyclin
- 2) Macrolide

Decrease P acne →
decrease lipase.
Decrease chemotactic
factor
Decrease neutophilic
migration

Retinoids

Indications →

severe acne
Moderate not respond
Gram -ve
Fulminant &
conglobate
0.5-1 mg

Anti-androgen

Cyproterone acetate
Flutamide
spironolactone

Others

Metformin
Vit A
Zinc
NSAID
Clofazimin
Prednisone
Dapson

COMMONLY USED COMBINED ORAL CONTRACEPTIVES	
Oral contraceptive®	Estrogen mcg/Progestin mcg
<i>FDA-approved for acne vulgaris</i>	
Ortho Tri-Cyclen	Ethinyl estradiol 35/norgestimate 180, 215, 250
Ethrostep	Ethinyl estradiol 20, 30, 35/norethindrone 1000
Yaz, Loryna, Nikki, Beyaz*	Ethinyl estradiol 20/drospirenone 3000
<i>Clinical data to support use</i>	
Alesse	Ethinyl estradiol 20/levonorgestrel 100
Diane-35†	Ethinyl estradiol 35/cyproterone acetate 2000
Yasmin, Syeda, Yaela, Safyral*	Ethinyl estradiol 30/drospirenone 3000
Natazia	Estradiol valerate 1000, 2000, 3000/dienogest 2000, 3000



Fig. 36.18 Cutaneous complications of isotretinoin therapy. **A** Impetigo in a patient treated with isotretinoin. Multiple serous crusts are evident. **B** Pyogenic granuloma-like healing on the chest of an adolescent boy on his third month of isotretinoin therapy. This is more likely to occur when isotretinoin therapy is started at a full dose rather than a lower initial dose, especially in teenage boys.



Other modalities

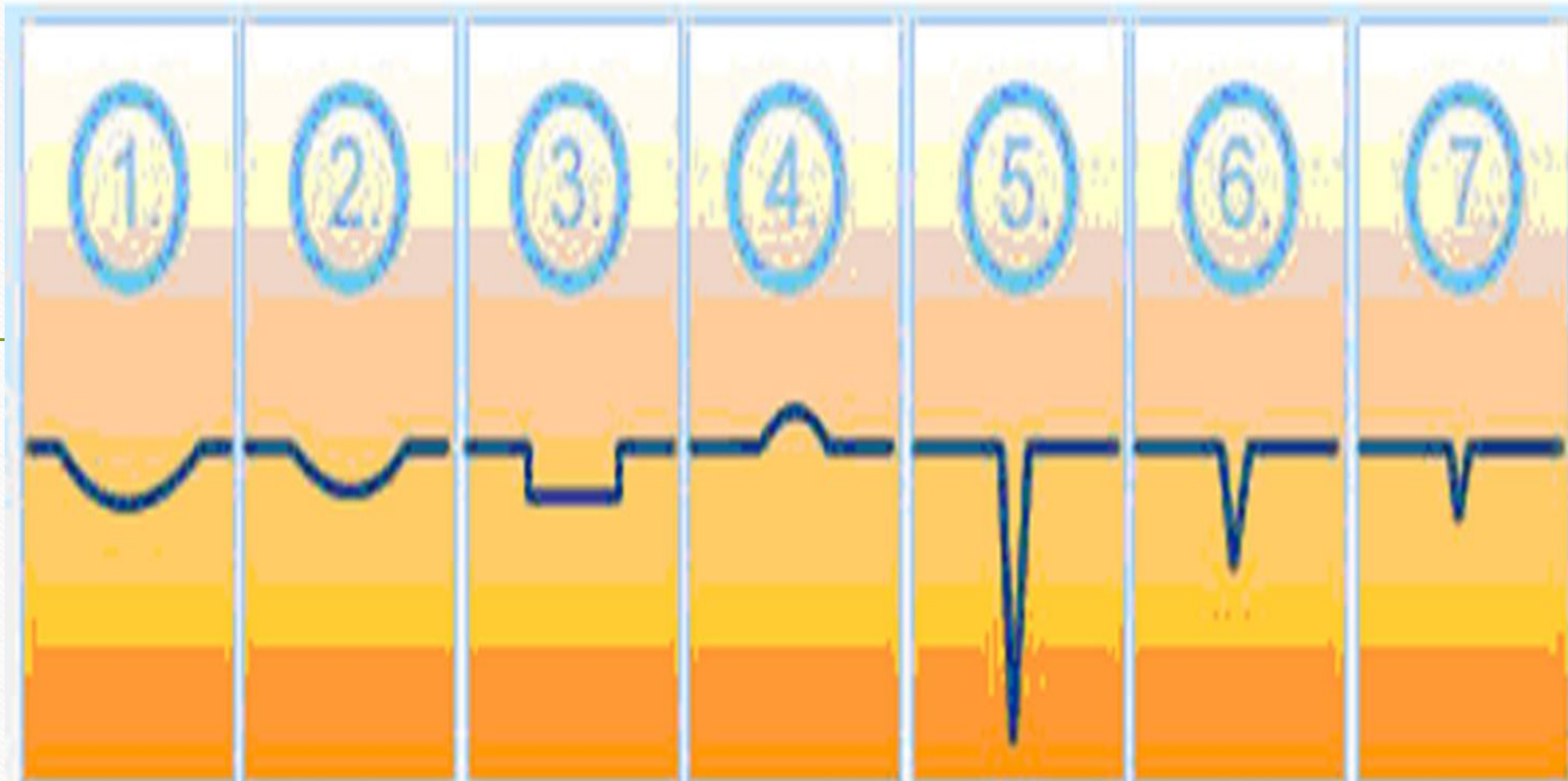
Physical
Light-comedonal extraction-
cryotherapy

Ttt of scar
Derma roller
Cross technique
PRP
Co2 laser

Scar formation

Hypertrophic scar
keloid

Atrophic scar
1] ice pick scar
2] box car scar
3] rolling scar



Deep
Rolling
Scar

Shallow
Rolling
(Saucer)

Box
Car

Raised
Papule
(Bump)

Deep
Pit

Shallow
Pit

Enlarged
Pore

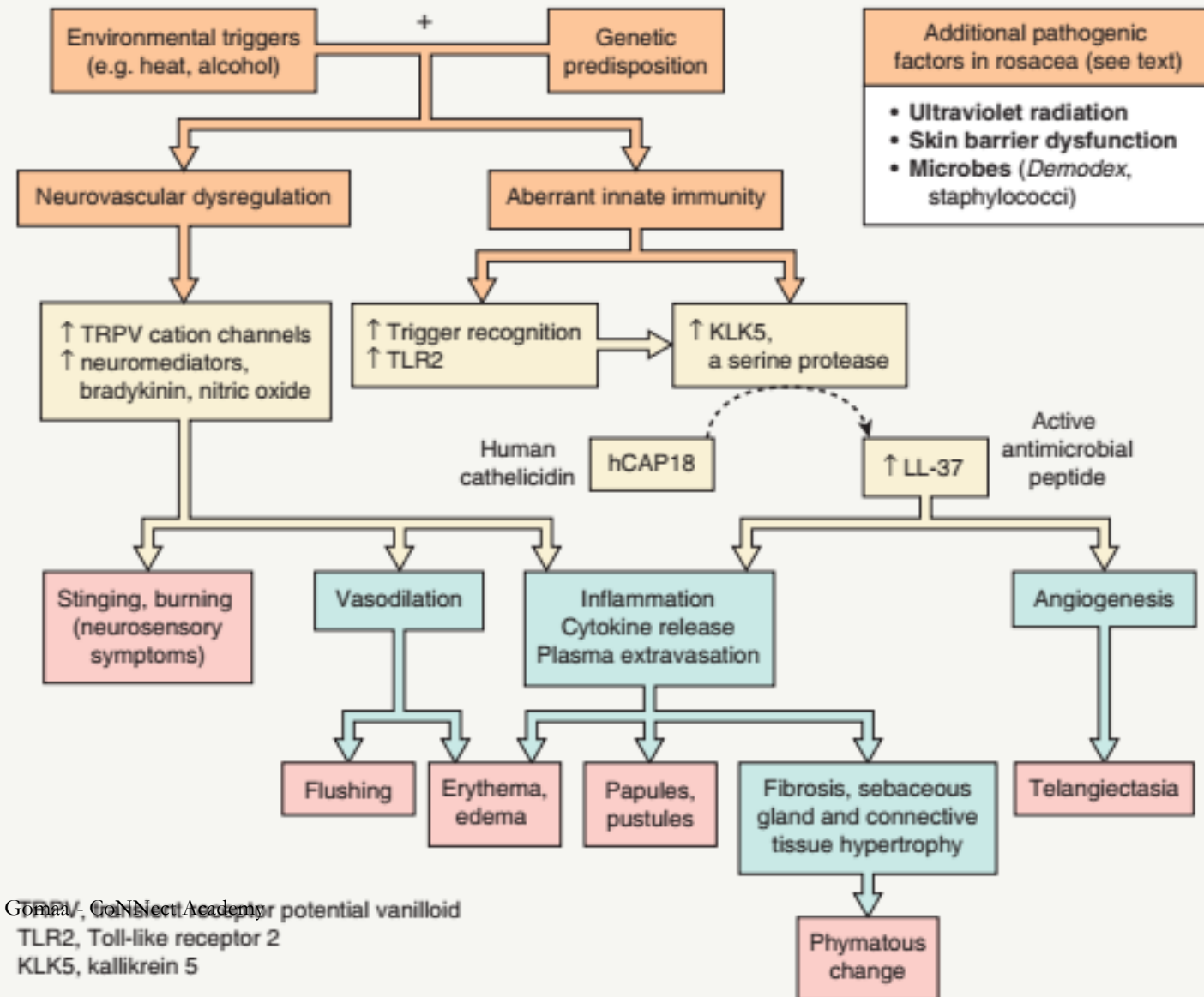


Fig. 36.9 "Ice-pick" scarring secondary to acne.

rosacea

- Def:- chronic vascular inflammatory disorder limited to central phase characterized by facial erythema, telagectesia, papules & pustules
- Epidemiology :- female 30-50 ys / skin type 1&2
- C/pic :- site central phase → cheeks, skin, forehead & nose

MAJOR PATHOMECHANISMS IN ROSACEA



Features of rosacea

- 1ry :- flushing, non transient erythema, telangiectasia, papules & pustules.
- 2ry: burning, stinging, dry appearance

subtypes

Erythematotelangiectatic

Papulo-bustular

Phymatous

ocular

Flushing
Telengectasia
Edema

Inflammatory rosacea
No scar
No comedon

Red deep regular lobulated thickening of skin of nose + follicular dilatation

Difficult to diagnose
Can occur without skin



Fig. 37.2 Erythematotelangiectatic rosacea. Persistent erythema of the medial and lateral cheeks is seen. In this patient, there are no telangiectasias, indicating mild (grade 1) disease.



Fig. 37.3 Moderate papulopustular rosacea of the forehead. Note the



Fig. 37.4 Moderate to severe papulopustular rosacea. There is a typical centrofacial distribution of erythema, telangiectasias, papules and pustules. In



Fig. 37.6 Rhinophyma. Hypertrophy of sebaceous glands and connective tissue as well as patulous follicles are seen. The changes are more prominent in

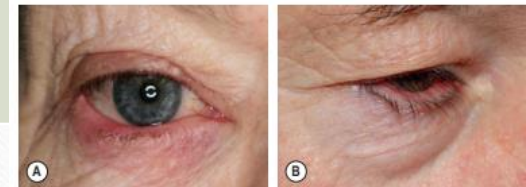


Fig. 37.9 Ocular rosacea. A Tiny concretions of keratin (conical dandruff) are visible at the bases of some of the eyelashes of the lower eyelid. There is also evidence of blepharitis of the lower eyelid and conjunctival injection. B Erythema of the mucosal portion of the lower eyelid and ectropion. C Marked injection of the conjunctivae, leading to the appearance of red eyes. Ectropion is also present.

variants

granulomatous

More persistent
Discrete
Red brown
Facial papule & nodule



Fig. 37.10
Granulomatous rosacea, lupus miliaris disseminatus faciei variant. Monomorphic, discrete skin-colored to brown papules scattered on the face that are more persistent than the lesions of papulopustular rosacea. Histologically, granulomas with central caseation necrosis were observed, which led to the diagnosis of lupus miliaris disseminatus faciei.

Pyoderma facii

Rosacea fulminant
Post adolescent women
20-40 years with no history of acne suddenly erupted purulent nodulocystic lesion in face
No systemic
No comedon
No back & chest
Ttt → tetracyclin & steroid



Fig. 37.16 Morbihan disease. Erythematous, firm, non-pitting, asymptomatic swelling of the upper midface. Areas of greatest involvement can acquire a



eFig. 37.2 Distortion of nasal tissue due to nodular hypertrophy of sebaceous glands. Electrosurgery or laser therapy can be used to debulk and resculpt this nose.

CLASSIFICATION, CLINICAL FEATURES AND MANAGEMENT OF SUBTYPES OF ROSACEA				
Subtype	Predominant clinical features	Severity		Management
		Grade	Features	
Erythematotelangiectatic (ETTR)	• Persistent centrofacial erythema • Flushing • Telangiectasias • Skin sensitivity	1	Occasional mild flushing; faint persistent centrofacial erythema; few telangiectasias	• General recommendations for facial skin care (see Table 37.4) • Reduce flushing • Minimize irritation • Topical treatments used in papulopustular rosacea may cause irritation • Laser therapy of prominent blood vessels in grades 2 and 3 disease
		2	Frequent troublesome flushing; moderate persistent centrofacial erythema; several distinct telangiectasias	
		3	Frequent severe flushing; pronounced persistent centrofacial erythema; many prominent telangiectasias; possible edema	
Papulopustular (PPR)	• Persistent centrofacial erythema • Papules • Pustules/papulopustules • Overlap with other subtypes may occur	1	Few papules and/or papulopustules; mild persistent centrofacial erythema	• Topical or systemic medications for grades 1 and 2 disease • Systemic medications for grade 3 • May require repeated courses or gradual taper
		2	Several papules and/or papulopustules; moderate persistent centrofacial erythema	
		3	Extensive papules and/or papulopustules; pronounced persistent centrofacial erythema; inflammatory plaques or edema may be present	
Phymatous	• Thickened, nodular skin • Prominent pores • Can affect nose (rhinophyma), chin (gnathophyma), forehead (metophyma), ears (otophyma), eyelids (blepharophyma) • May be associated with other features of rosacea or occur in isolation	1	Puffiness; mildly patulous follicles; no clinically apparent hypertrophy of connective tissue or sebaceous glands; no change in contour	• Surgical or laser therapy for grades 2 and 3 rhinophyma • Other phymas are more difficult to treat but may improve with systemic treatments, if an inflammatory component is present
		2	Moderate swelling; moderately dilated patulous follicles; clinically, mild hypertrophy of the sebaceous glands or connective tissue; change in nasal contour without nodular component	
		3	Marked swelling; large dilated follicles; distortion of contour due to hypertrophy of the sebaceous glands and/or connective tissue, with a nodular component	
Ocular	• Dry, gritty sensation • Blepharitis • Conjunctivitis • Chalazia and hordeola • Keratitis, episcleritis, scleritis, iritis (rare)	1	Mild itch, dryness or grittiness; fine scaling of eyelid margins; telangiectasia of eyelid margins; mild conjunctival injection	• Topical medications for grade 1 • Systemic medications for grade 2 • Refer to ophthalmologist for persistent grade 1 or 2 disease or suspected grade 3 disease
		2	Burning or stinging; crusting, irregularity, erythema and/or edema of eyelid margins; definite conjunctival injection; chalazion or hordeolum	
		3	Pain, photosensitivity, blurred vision; severe eyelid changes with loss of lashes, severe conjunctival inflammation; corneal changes, with potential loss of vision; episcleritis, scleritis; iritis	

GENERAL RECOMMENDATIONS FOR FACIAL SKIN CARE AND EDUCATION IN PATIENTS WITH ROSACEA

Facial skin care

- Wash with lukewarm water and use soap-free cleansers that are pH balanced
- Cleansers are applied gently with fingertips
- Use sunscreens with both UVA and UVB protection and an SPF ≥ 30
- Sunscreens containing the inorganic filters titanium dioxide and/or zinc oxide are usually well tolerated
- Use cosmetics and sunscreens that contain protective silicones
- Water-soluble facial powder containing inert green pigment helps to neutralize the perception of erythema
- Moisturizers containing humectants (e.g. glycerin) and occlusives (e.g. petrolatum) help to repair the epidermal barrier
- Avoid astringents, toners, and abrasive exfoliators
- Avoid cosmetics that contain alcohol, menthols, camphor, witch hazel, fragrance, peppermint, and eucalyptus oil
- Avoid waterproof cosmetics and heavy foundations that are difficult to remove without irritating solvents or physical scrubbing
- Avoid procedures such as glycolic peels or dermabrasion

MEDICAL AND SURGICAL THERAPIES FOR ROSACEA

Treatment	Comments and/or Doses
Erythematotelangiectatic	
Facial skin care recommendations (see Table 37.4)	Particularly useful as this subtype is prone to skin irritation and “sensitivity”
Photoprotection	UVR may potentiate dermal matrix damage
Topical agents, e.g. azelaic acid, metronidazole	May reduce erythema, but their use is often limited by their irritant effects
Topical brimonidine tartrate (0.33% gel)*†	Selective α_2 -adrenergic agonist that improves erythema
Topical oxymetazoline HCl (1% cream)*	Selective α_{1A} -adrenergic agonist that improves erythema
Laser therapy†	Use of vascular lasers (e.g. pulsed dye, potassium titanyl phosphate) as well as intense pulsed light may improve grades 2 and 3

Papulopustular	
Topical	
Metronidazole (0.75% gel or cream; 1% cream)* [†] once or twice daily	Can be used as initial treatment to clear inflammatory lesions or as indefinite maintenance therapy
Ivermectin (1% cream)* [†] once daily	More effective than placebo and slightly more effective than topical metronidazole in randomized controlled trials
Azeleic acid (15% gel)* [†] twice daily	Appears to be more effective than topical metronidazole but with more side effects, e.g. irritation Azeleic acid (20% cream twice daily) is a non-FDA-approved alternative dose
Sodium sulfacetamide (10%) and sulphur (5%) in a cream or lotion* once or twice daily	May include 10% urea
Erythromycin (2% solution) twice daily	Alcohol in solution may reduce tolerance
Clindamycin (1% lotion) daily	
Benzoyl peroxide 5% plus clindamycin 1% daily	May cause skin irritation
Tretinoin (0.025% cream; 0.05% cream; 0.01% gel) daily	Alters epidermal keratinization and may improve photodamage Poorly tolerated by some patients
Permethrin (5% cream) daily–weekly	Shown to be as effective as topical metronidazole for the treatment of papules and erythema May have future role in combination with antibiotics, but further studies needed
Pimecrolimus (1% cream) or tacrolimus (0.03%, 0.1% ointment) twice daily	Some studies have shown improvement in erythema, but there have been case reports of exacerbations, so further studies needed
Systemic	
Doxycycline* [†]	40 mg daily (30 mg immediate release and 10 mg delayed release) for 4–8 weeks As effective as the 100 mg dose but with fewer adverse effects
Doxycycline	50–100 mg once or twice daily for 4–8 weeks
Minocycline	50–100 mg twice daily or sustained action formula (1 mg/kg) daily for 4–8 weeks**
Tetracycline [†]	250–500 mg twice daily for 4–8 weeks
Erythromycin	250–500 mg once or twice daily for 4–8 weeks
Azithromycin	250–500 mg (5–10 mg/kg) thrice weekly for 4–8 weeks
Metronidazole	200 mg once or twice daily for 4–8 weeks
Isotretinoin [†]	0.3 mg/kg/day

Phymatous

Isotretinoin	May reduce nasal volume and halt the progression of rhinophyma
Surgical excision	Can effectively debulk and resculpt the nose
Electrosurgery	
CO ₂ laser	

Ocular

Eyelid hygiene and artificial tears	Frequently used to treat mild disease
Fusidic acid	Useful for maintaining remission following treatment of grades 2 and 3 disease with systemic antibiotics
Metronidazole gel	
Cyclosporine 0.5% ophthalmic emulsion†	More effective than artificial tears in treatment of ocular rosacea
Systemic antibiotics (see above section)	For grade 2–3 disease

